

EXPERIMENT 1 — WINDOWS DEFENDER — MALWARE DETECTION AND SYSTEM SECURITY

Aim

To study the working of Windows Defender / Microsoft Defender Antivirus and perform different types of malware scans to detect and protect the system from security threats.

Description

Windows Defender, currently known as Microsoft Defender Antivirus, is a built-in security component of the Windows operating system. It provides protection against malware such as viruses, spyware, trojans, ransomware and other malicious software.

It provides real-time protection by continuously monitoring files, applications and system activities. When a suspicious file or activity is detected, Defender can block, quarantine or remove the threat.

Requirements

Category	Requirement
Operating System	Windows operating system
Security Tool	Windows Defender / Microsoft Defender Antivirus
Interface	Windows Security application
Command Line	Windows PowerShell

PROCEDURE

A. Update Windows Defender

1. Open the Windows Security application.
2. Navigate to **Virus & threat protection**.
3. Check the available protection and security information.
4. Use **Check for updates** to obtain the latest available security intelligence updates.
5. After updating, continue with the required malware scan.

B. Quick Scan

1. Open Windows Security.
2. Select **Virus & threat protection**.
3. Select **Quick scan**.
4. Allow Windows Defender to scan the system.
5. Wait until the scan is completed.
6. Check the scan result and Protection History for any detected threats and actions taken.

A Quick Scan takes less time because it concentrates on common malware locations. It is useful for obtaining a relatively quick security check of the system.

C. Full Scan

1. Open **Virus & threat protection**.
2. Select **Scan options**.
3. Choose **Full scan**.

4. Start the scan and allow it to examine the system.
5. Wait for the scan to complete and examine the result.

A Full Scan performs a more comprehensive examination of the system compared with a Quick Scan.

D. Custom Scan

1. Open **Virus & threat protection** and select **Scan options**.
2. Select **Custom scan**.
3. Choose the required file or folder.
4. Start the scan.
5. Examine the result for detected threats or other security information.

E. PowerShell Operations

The following PowerShell commands can be used to obtain Defender status, initiate scans, update security intelligence and view detected threats:

```
Get-MpComputerStatus  
Start-MpScan -ScanType QuickScan  
Start-MpScan -ScanType FullScan  
Update-MpSignature  
Get-MpThreatDetection
```

RESULT ANALYSIS

Windows Defender successfully checks the system for potentially malicious files and activities. The experiment demonstrates that different scan types can be selected according to the required level and scope of examination.

A Quick Scan takes less time because it concentrates on common malware locations, while a Full Scan performs a more comprehensive examination of the system. A Custom Scan can be used when a particular file or folder needs to be checked.

The Protection History provides information about threats detected and the actions taken. PowerShell provides another method for obtaining Defender status and initiating scans.

The experiment demonstrates the importance of endpoint security because malicious software can enter a computer through downloads, email attachments, removable storage, compromised applications and malicious websites.

RESULT

Windows Defender was successfully used to study system protection and perform malware scans. Quick Scan, Full Scan and Custom Scan provide different approaches for checking the system, while PowerShell commands provide command-line methods for status checking, scanning, updating security intelligence and viewing detected threats.

CONCLUSION

The experiment demonstrates the working of Microsoft Defender Antivirus as an endpoint-security component. Regular updates and appropriate scanning methods help detect and protect the system from malware and other security threats.

OBSERVATIONS

1. Windows Defender is a built-in security component of Windows.
2. Microsoft Defender Antivirus provides protection against different types of malware.
3. Real-time protection continuously monitors files, applications and system activities.
4. A suspicious file or activity may be blocked, quarantined or removed.
5. Virus & threat protection provides access to malware scanning features.
6. Security intelligence updates can be checked through the update option.

7. Quick Scan concentrates on common malware locations.
8. Quick Scan generally requires less time than a Full Scan.
9. Full Scan performs a more comprehensive examination of the system.
10. Custom Scan allows a selected file or folder to be checked.
11. Protection History provides information about detected threats.
12. Protection History also records actions taken for detected threats.
13. PowerShell can be used to obtain Microsoft Defender status.
14. Start-MpScan can be used to initiate scans from PowerShell.
15. Update-MpSignature can be used to update Defender security intelligence.
16. Get-MpThreatDetection can be used to view detected threats.
17. Different scan types can be selected according to the required scope.
18. Malware may enter a system through downloads, email attachments or removable storage.
19. Compromised applications and malicious websites are also possible sources of malware.
20. The experiment demonstrates the importance of endpoint security.

Practical Flow: Open Windows Security → Virus & threat protection → Check for updates → Select scan type → Run scan → Examine result/Protection History. PowerShell alternative: Check status → Update signatures → Start scan → View detected threats.